

Cardano - Ecosystem Stack

Cardano

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About OpenZeppelin

Founded in 2015, OpenZeppelin is the world leader in securing blockchain applications and smart contracts. Its bedrock open source Contract Libraries are a public good and industry standard for smart contract development. OpenZeppelin's professional expertise, together with its [Open Source Stack](#), integrates through clients' development lifecycles, so teams can plan, code, audit, deploy and operate projects faster and more safely.

OpenZeppelin is trusted by the world's leading projects:



coinbase

AAVE



OPTIMISM



Explore our [Open Source Impact](#) across the industry.

Relevant metrics

+\$36 trillion

Total Value Transferred via OpenZeppelin Contracts

Explore our [Dune Data Dashboard](#) ↗

\$110B+

In TVL importing
OpenZeppelin Contracts

Market Share of ~85%

OpenZeppelin Contracts in DeFi
TVL

5.9B+

Transactions processed since
2018

Off-chain statistics

~590K

Average Weekly NPM
Downloads

+1.8M Yearly Views*

On the OpenZeppelin Docs

+1.6M Yearly Views*

On the Contracts Wizard

84% market penetration

for OpenZeppelin Contracts across top 50 DeFi protocols

*in 2025

Abstract

This proposal requests funding for the design, implementation, security audit, and ecosystem adoption of a comprehensive open-source development stack for Cardano, delivered over an **initial 12 months in quarterly milestones**.

It bundles three tightly coupled workstreams into a single, integrated proposal, led by the Reference Implementations and built around them:



1. OpenZeppelin Reference Implementations for Cardano - three production-ready, end-to-end blueprints targeting strategic DeFi and institutional use cases (a **Cardano Liquid Staking Protocol**, a **Self-Repaying Loans Protocol**, and a **Tokenized Money Market Fund**), each including working reference code, architecture documentation, a demo front-end, and a threat model.



2. OpenZeppelin Contracts Library for Cardano - a secure, audited, standardized contracts library tailored for Cardano's eUTXO model and Plutus environment, serving the same role OpenZeppelin Contracts serve for Solidity. Its roadmap feeds directly from the needs and use cases that arise from the Reference Implementations, so **every primitive is battle-tested in a real application before release**. Includes Contracts Wizard integration, AI development tools, and documentation.



3. Security Retainer - dedicated security capacity covering line-by-line audits of the library and full-stack security reviews and penetration testing of every Reference Implementation, ensuring each release meets OpenZeppelin's established security standards.

Additionally, OpenZeppelin commits to a **Developer Enablement and Co-Marketing program**, included at no additional cost, to drive ecosystem and institutional adoption of the libraries and Reference Implementations.

Motivation

The Opportunity

Cardano is at an inflection point. With Circle's [USDCx now live](#), LayerZero integration underway, and growing institutional interest in what eUTXO can uniquely offer, the ecosystem is consolidating a solid foundation to scale. **What's missing are the trusted, consolidated building blocks that enable developers to build securely, and the clear application paths that make Cardano the obvious choice for protocols and institutions evaluating where to deploy next.**

The Problem

- **No reference implementations** for the [composable financial products Cardano's ecosystem needs](#): no liquid staking, no self-repaying loans, no composable DeFi building blocks.
- **No standardized, audited contract libraries.** Audited code is scattered across Cardano Foundation repositories, Intersect resources, and individual project codebases, with no single, authoritative contracts library that builders can import and compose with confidence. Each team independently rebuilds the same foundational components, slowing adoption, fragmenting the ecosystem, and increasing security risk.

Why OpenZeppelin

Founded in 2015, OpenZeppelin is the world leader in securing blockchain applications and smart contracts, and its open-source Contract Libraries are the industry standard for smart contract development.

Metric	Value
Total Value Transferred via OpenZeppelin Contracts	+\$36 trillion
TVL importing OpenZeppelin Contracts	\$110B+
Market share in top 50 DeFi protocols	~85%
Transactions processed since 2018	5.9B+
Average weekly NPM downloads	~590K
Yearly documentation views	+1.8M
Yearly Contracts Wizard views	+1.6M

OpenZeppelin has built trusted contract libraries and developer tools across multiple ecosystems, each time becoming the standard foundation developers build on:

Ecosystem	Scope
Ethereum	Solidity Contracts (10+ years), EIPs/ERCs standards, gold-standard audits, incident response
Starknet	Contracts Library for Cairo, SNIPs standards, security tooling, Contracts Wizard
Arbitrum	Contracts Library for Stylus (Rust), AIPs standards, Security Council member, Contracts Wizard
Stellar	Contracts Library for Soroban, SEPs standards, Ecosystem Stack, Contracts Wizard
Midnight	Contracts Library for Compact, LunarSwap reference DEX, privacy-preserving UTXO architecture
Uniswap	Contracts Library for Hooks, gold-standard audits for v4, Contracts Wizard
Polkadot	Library of runtime templates for Substrate, PSPs standards
Zama	Confidential Contracts Library for Zama's fhEVM

Our **Midnight engagement is particularly relevant**: we built [LunarSwap](#), a working reference DEX on UTXO-based architecture, solving many of the same challenges Cardano's eUTXO and Plutus environment presents.

As the [primary security partner for financial institutions](#) including **DTCC and WisdomTree**, OpenZeppelin is uniquely positioned to help Cardano accelerate composability across TradFi and DeFi.

Strategic Fit

This is not a cold start. OpenZeppelin has built deep expertise in UTXO-based architectures through our Midnight engagement, and our team brings a decade of experience translating standards into production-ready building blocks across ecosystems. **Our approach complements the ecosystem's needs: we build trusted open-source building blocks that accelerate ecosystem adoption while collaborating closely with Input Output, the Cardano Foundation, Intersect and other ecosystem stakeholders.** As our team builds on Cardano's stack, we will provide developer-experience feedback on existing ecosystem tooling (developer frameworks, SDKs, Aiken tooling) and contribute improvements and security input that benefit the entire ecosystem, helping streamline the secure development path for all builders.

Specification

1. Objective

Deliver shared, open-source ecosystem infrastructure for Cardano that:

- **Accelerates DeFi and Institutional adoption** by providing production-ready Reference Implementations that teams can evaluate, fork, and build upon rather than starting from scratch.
- **Standardizes secure development** by providing audited, composable contract primitives that developers import and compose, reducing fragmentation, lowering security risk, and accelerating time-to-market.
- **Bridges the developer gap** by delivering the same streamlined developer experience (Contracts Library, Contracts Wizard, AI tools, documentation) that has driven ~85% market penetration in Ethereum.
- **Improves ecosystem tooling** through active collaboration: as our team builds on Cardano's stack, we provide developer-experience feedback and contribute improvements that benefit the entire ecosystem.

2. Implementation Mechanics

How the Workstreams Connect

The three workstreams are **integrated into a single proposal** to accelerate critical ecosystem DeFi and financial use cases while ensuring secure and modular component composability.

The Reference Implementations and the Contracts Library are designed to reinforce each other. **The library roadmap feeds directly from the needs and use cases that arise from the Reference Implementations: as we build each RI, we identify and prioritize the reusable contract primitives required, extract and generalize them into standalone library components, and then use those library components as the building blocks within the Reference Implementation itself.** This means every library component is validated in a real reference application before release, and every Reference Implementation showcases integrations of our secure contract standard implementations while documenting usage of the best ecosystem tools and off-chain components.

By working on strategic DeFi and Institutional use cases for Reference Implementations in parallel with the library, we continuously adjust our library roadmap to bring the smart contract primitives

that will accelerate DeFi and TradFi teams. At the same time, we provide active feedback to improve any tools and solutions integrated into our Reference Implementations.

Security is the baseline that makes the Reference Implementations and contract libraries production grade and ready for adoption.

Workstream A: Reference Implementations

OpenZeppelin Reference Implementations are the next evolution of OpenZeppelin standards. They are specifically designed to target critical bottlenecks that both DeFi teams and financial institutions face when trying to build on or integrate with Cardano. They dramatically compress timelines by providing production-ready blueprints and end-to-end working examples that enable teams to evaluate, decide, and deploy using battle-tested patterns rather than starting from scratch.

Each end-to-end implementation will include:

- **Working Reference Code** - Production-ready implementations integrating with Cardano for particular use cases, with all on-chain and off-chain elements assembled into a single packaged offering, including deployment utilities for immediate evaluation on testnets (Preview/Preprod).
- **Architecture Documentation** - Component maps showing integration patterns with custody, compliance, and risk management systems.
- **Demo Front-End** - Prototype end-to-end UI supporting core flows and technical evaluation, with reusable white-label components.
- **Threat Model and Security Considerations** - Risk assessment specific to each use case, addressing attack vectors and failure modes particular to the eUTXO model. All on-chain outputs undergo security audits (covered under Workstream C).

All Reference Implementations will be **designed with institutional composability in mind**, and can include patterns such as credential-based access and multi-signature controls. This way, the implementations can serve both a crypto-native team looking to build or port their solution to Cardano, and a regulated institution that wants to provide DeFi utility on top of their tokenized assets or participate in Cardano's growing stablecoin ecosystem. As needed, Reference Implementations will also include **FI Evaluation Guides** tailored to accelerate compliance and procurement for financial institutions.

Reference Implementations (preliminary; specific use cases to be defined through joint scoping sessions with Input Output, the Cardano Foundation, and ecosystem partners):

1. **Cardano Liquid Staking Protocol** - End-to-end blueprint for an ADA liquid staking protocol that issues a transferable, yield-bearing token representing staked ADA. ADA delegation is non-custodial, but when ADA moves into a smart contract pool the user typically loses individual delegation choice, and pooled ADA is frequently left undelegated or only partially delegated, with rewards accruing to the protocol or pool rather than to users' chosen SPOs. No neutral, open-source, audited LST primitive exists that new dApps, L2s, or sidechains can adopt as a default. A Lido-style Liquid Staking Token (LST) reference implementation closes that gap: users deposit ADA, the protocol delegates across an SPO set, and users receive an LST that accrues value as staking rewards arrive, composable as collateral and as a swap asset. The Reference Implementation will serve as a building block for other Reference Implementations, restaking and downstream protocols that need a yield-bearing ADA primitive.

The implementation composes standardized, audited primitives from our Contracts Library (e.g., Vaults, Access Control, Timelock, Staking and Delegation Utilities) into a production-ready protocol rather than rebuilding them from scratch, while its requirements feed back into and shape the library roadmap.

2. **Self-Repaying Loans Protocol** - End-to-end blueprint for a self-repaying loan protocol on Cardano, inspired by Alchemix on Ethereum: users deposit yield-bearing collateral and borrow against it at a fixed maximum loan-to-value ratio with no interest. The implementation could cover deposit and synthetic-debt issuance, a periodic harvest mechanism that applies accrued yield against outstanding user debt, and a redemption pathway. The protocol will build on the Vaults, DeFi Math, Access Control, Governance, and Timelock primitives from our Contracts Library, and may compose with Cardano's existing synthetic-asset and lending infrastructure where useful. No Cardano protocol today auto-applies yield from collateral to debt, this self-repaying mechanic remains an unfilled primitive, and the design pairs naturally with the Liquid Staking Reference Implementation as its highest-leverage collateral source.
3. **Tokenized Money Market Fund** - End-to-end blueprint for a tokenized money market fund on Cardano, enabling asset managers and institutions to issue and manage regulated fund shares. The reference implementation will demonstrate the full lifecycle of a tokenized fund, from share issuance with KYC/AML credential verification, through NAV-based subscription and redemption flows, to dividend distribution and regulatory reporting. The implementation will leverage the Vaults, Credentials and Claims, and Access Control primitives from our Contracts Library, and integrate with Cardano's stablecoin infrastructure including Circle's USDCx for settlement. This Reference Implementation directly addresses Cardano's ambition to attract institutional participants by providing a production-ready path for one of the fastest-growing use cases in tokenized finance, an

area where BlackRock, Franklin Templeton, and others are already deploying billions on other chains. By demonstrating that Cardano's eUTXO model can support institutional-grade asset management with built-in compliance, this blueprint serves as a powerful answer to the question of why an institution would choose Cardano.

Potential alternative or additional directions for future Reference Implementations include tokenized deposits, real world asset tokenization, cross-chain stablecoin payment orchestration, stablecoin aggregation, onchain insurance, and other DeFi applications with institutional compliance controls. **The roadmap is flexible and will be adjusted based on ecosystem priorities and stakeholder feedback throughout the engagement.**

Open Source Tooling Integration and Ecosystem Collaboration

By working on strategic DeFi use cases for Reference Implementations, OpenZeppelin will actively use, provide feedback on, and integrate with ecosystem tooling and infrastructure - existing developer frameworks, SDKs, and off-chain tools the Cardano community has built, as well as new developments needed in the Reference Implementations. OpenZeppelin will also integrate tools from the [OpenZeppelin Stack](#) into Cardano as needed for the Reference Implementations, such as [Monitor](#) for on-chain event detection and automated alerting, or [Relayer](#) for programmable transaction execution enabling sponsored-transaction patterns. As our team gains deep familiarity with Cardano's architecture and tooling landscape, we will contribute improvements and security input that benefit the entire ecosystem.

Workstream B: Contracts Library

OpenZeppelin proposes a secure, audited, and well-documented contracts library for Cardano that serves the same role as OpenZeppelin Contracts for Solidity. Cardano today has active smart contract development across multiple languages, from Haskell-based PlutusTx to the increasingly adopted Aiken, but **there is no single, authoritative contracts library** that developers can import and compose with confidence. **The specific smart contract language targeted will be determined during the initial evaluation phase, in collaboration with Input Output and ecosystem stakeholders, based on developer adoption trends and Cardano's technical roadmap.**

Because fungible and non-fungible tokens on Cardano are issued natively by the ledger as Cardano Native Tokens (CNTs), the OpenZeppelin Contracts library for Cardano will focus on the **application-layer primitives** (access control, governance, vaults, cryptography, DeFi building blocks, and account abstraction) that operate over native assets to enable secure, composable dApp development.

Library Support for Existing CIPs

OpenZeppelin will add library support for existing CIPs. The criteria to prioritize these will be based both on strategic ecosystem needs and those primitives that will be relevant and utilized in our Reference Implementations. Having developed standards and libraries covering tokens, stablecoins, application frameworks, and interoperability across Ethereum, Stellar, Starknet, and other ecosystems, we bring deep experience in translating these concepts into secure, production-ready building blocks.

- **CIP-113 - Programmable Token Standard.** CIP-113 introduces programmable/permissioned token behavior, in the same family as Ethereum's ERC-1404 and ERC-3643 (T-REX). OpenZeppelin will contribute to the standard's evolution and ensure that, once the standard is stable and ecosystem-ready, the resulting library implementation is built on solid foundations.
- **Additional CIPs** will be evaluated and prioritized based on concrete ecosystem needs and adoption signals. Where the community sees value, OpenZeppelin will author or co-author new CIPs and shepherd them through the standards process.

Additional Library Components

OpenZeppelin will provide additional library components for other priority features for Cardano, so that these features may be used within the RIs and by developers for other applications.

OpenZeppelin will also submit new CIPs associated with these, as deemed appropriate and desirable by the community, to promote standardization and reuse. Smart contract features and standards covered may include:

- **Access Control & Roles** - Foundational permissioning patterns, single-owner, role-based access control, and policy-based access manager equivalents; adapted for Cardano's native multi-signature capabilities and credential system. Access control is one of the most widely-imported families of contracts in OpenZeppelin's Solidity library and underpins virtually every protocol, both crypto-native and institutional.
- **Cryptographic Utilities** - Merkle proof verification, signature checking, message hashing, ECDSA recovery, and EIP-712-equivalent typed-data signing patterns adapted for Cardano's signing schemes. These primitives are the substrate for airdrops, oracle integrations, off-chain authorization, allowlists, and credential verification, and rank as one of the most universally-used components of the Solidity library.
- **DeFi Math Library** - With safe math primitives and arithmetic utilities tailored to common DeFi use cases on eUTXO, including fixed-point arithmetic and price computation patterns.
- **Vaults** - Core to lending, permissioned and permissionless, multi-asset, and multi-chain. Adapted for Cardano's eUTXO model where vault state is managed through UTXO consumption rather than account storage, drawing on our experience implementing tokenized vault standards across Ethereum (ERC-4626), Stellar (SEP-56), and Starknet.
- **DeFi Primitives** - DeFi building blocks adapted for eUTXO, including liquidity pool management, swap logic, fee structures, and modular hook patterns for pool customization (see [Uniswap Hooks](#)). Designed to work with the concurrency patterns unique to Cardano, where multiple users interacting with the same pool requires careful UTXO management.
- **Timelock and Pause** - Useful for security controls, governance delays, and emergency circuit breakers in DeFi protocols.
- **Staking and Delegation Utilities** - Leveraging Cardano's unique liquid staking model where delegation does not require locking, enabling DeFi protocols to preserve delegation rights while assets are deployed in lending or liquidity pools.
- **Governance** - Token-voted governance components (Governor, Votes, Delegation, Quorum, and Timelock execution) enabling DApp-level governance on Cardano. **Cardano lacks a standardized, audited, dApp governance standard with the modularity developers expect from other ecosystems.** The Governance module closes that gap and could integrate with Cardano's protocol-level governance (CIP-1694, DRep voting, treasury)

where the two layers compose. OpenZeppelin authored the Governor contracts used today by Arbitrum DAO, Compound, ENS, and most major Ethereum governance systems.

- **Credentials and Claims** - Necessary for compliance checks on DeFi participants, enabling regulated assets to enforce KYC/AML requirements through on-chain credential verification, aligned with supporting CIP-113 programmable token policies.
- **Modular Account Abstraction & Multi-Sig Accounts** - Modular multi-signature patterns necessary for DeFi participants (LPs and borrowers) and institutional custody requirements, leveraging Cardano's native multi-sig capabilities. This component family will also explore smart-account / account-abstraction patterns adapted for Cardano where the script and ledger semantics allow.
- **Vesting Contracts** - For token distribution schedules with configurable cliff and linear vesting periods.
- **Standardized Messaging Gateway** - Useful to add cross-chain support compatible with LayerZero and other interoperability providers, aligned with the ongoing LayerZero integration into Cardano's ecosystem.
- **Cardano Contract ABI** - Cardano lacks an Ethereum-style contract ABI; CIP-57 covers only datum and redeemer types, not the actions a contract supports. OpenZeppelin will design and propose a new CIP, and may include tooling that facilitates viewing the ABI for a contract.
- **Additional Cardano Standards** - To be determined throughout the engagement in collaboration with Input Output, Cardano Foundation, and ecosystem developers, for Cardano-specific use cases. Some possible use cases could include:
 - **Position Utilities** - Standards and utilities for transferable positions (loan positions, LP receipts, governance voting power), building on and extending CIP-68.
 - **Token Auctions** - For token launches and various auction formats.
 - **Upgradeable Contracts** - With built-in support for secure and maintainable upgrade paths.

Developer Tools

As part of the Cardano integration with OpenZeppelin, smart contract developers will benefit from the same streamlined experience they currently enjoy on Ethereum and other ecosystems partnered with OpenZeppelin.

- **Contracts Wizard and Documentation** - Developers bootstrap Cardano smart contract creation using a user-friendly interface and an AI assistant within the [OpenZeppelin Contracts Wizard](#). This integration includes code generation capabilities, templates, comprehensive usage guides, integration with ecosystem IDEs, and deployment workflows. With over +1.6M views in 2025, the OpenZeppelin Contracts Wizard will give thousands of

blockchain developers instant exposure to Cardano. The [OpenZeppelin UI Builder](#) will complement the Wizard by enabling developers to generate ready-to-deploy front-end interfaces for their Cardano smart contracts.

- **AI-Enhanced Development Tools** - [Contracts MCP Server](#) giving AI-assisted development environments such as Cursor, Claude, Gemini, Windsurf, and VS Code structured access to OpenZeppelin's library components. AI Development Skills encoding proven development workflows, library usage patterns, and security considerations. Claude Plugin packaging Cardano-specific MCP tools, skills, and development resources. This will include smart contract upgrade compatible enhancements (SCU).
- **Comprehensive Documentation:** A dedicated Cardano section on [OpenZeppelin Documentation](#) detailing contract types, interfaces, use cases, and best practices.

Workstream C: Security Retainer

OpenZeppelin will dedicate **22 researcher-weeks of security capacity over 12 months**, covering:

- **Smart Contract Security Audits:** Line-by-line review of all OpenZeppelin-produced smart contract code across the Contracts Library and Reference Implementations, identifying vulnerabilities, logic errors, and security flaws before each release.
- **Security Reviews:** Full-stack security assessments of each Reference Implementation, evaluating on-chain smart contracts alongside off-chain backend systems and front-end interfaces.
- **Penetration Testing:** Simulated real-world attacks against Reference Implementation deployments, covering application-level and infrastructure-level attack surfaces.
- **Continuous Coverage and AI-Security Agent:** Audits amplified by OpenZeppelin's AI security agent, integrated directly into the audit workflow as a supervised accelerator. The AI agent continuously analyzes the codebase, highlights high-risk patterns, validates invariants, and surfaces architectural risks before and between formal audit cycles, with all findings reviewed and validated by OpenZeppelin auditors.
- **Immunefi Bug Bounty:** Assistance to establish and manage a bug bounty program (OpenZeppelin will fund bounty payouts at the same levels as our standard OpenZeppelin Contracts bug bounty program).

This security capacity is scoped to OpenZeppelin-produced code. All security findings and reports will be published alongside the audited releases.

Developer Enablement and Co-Marketing (included at no cost)

To support the adoption milestones defined in this proposal, OpenZeppelin commits to the following at no additional cost to the fund:

- **Dedicated Technical Account Manager** for the Cardano Foundation.
- **Tailored technical content:** In-depth tutorials and step-by-step guides for each release.
- **Hackathon support:** Coaching and workshops at virtual and in-person events.
- **Interactive demos and feedback sessions:** Hands-on showcases followed by community Q&A.
- **Co-Marketing campaigns** for developer awareness and adoption of the Reference Implementations.
- **Dedicated Cardano Network Page** on OpenZeppelin's website (similar to [existing partner ecosystem pages](#)).
- **Technical workshops** at Cardano's flagship event and strategic conferences.
- **Quarterly community feedback sessions** via community calls, X-spaces, Discord, and developer office hours.

OpenZeppelin's reach: ~150K monthly documentation views, 58.5K X followers (~180K monthly impressions), 14.5K LinkedIn followers, ~13K monthly blog views.

3. Architectural Alignment

This proposal aligns with the Cardano ecosystem's core needs across multiple dimensions:

- **App Building and Developer Experience** - the Contracts Library, Contracts Wizard, AI tools, and documentation dramatically lower the barrier to building on Cardano.
- **Security and Resilience** - every library component and Reference Implementation undergoes rigorous security audit before release.
- **Financial Workflows and Composability:** All deliverables are designed for institutional composability with credential-based access, multi-signature controls and FI evaluation guides.
- **Ecosystem Standards Leadership** - OpenZeppelin contributes to CIP-113, proposes a Cardano Contract ABI, and brings expert leadership into ecosystem standards discussions.

4. Backward Compatibility

No backward compatibility impact. The Contracts Library and Reference Implementations are new additions to the ecosystem. Library components implementing existing CIPs will follow the published specifications and maintain compatibility with existing ecosystem tooling.

Milestones and Deliverables

Timelines assume an agreement is signed in **September 2026**, with ~20 days to resolve staffing and commence development from signing.

Milestone 1 - Foundations and dApp Framework (Q1: Oct 2026 → Dec 2026)

Focus: Foundational library components and Reference Implementation research

Reference Implementations:

- Research and Design for the 3 Reference Implementations
- Architecture Documents

Contracts Library:

- Access Control & Roles
- Cryptographic Utilities
- DeFi Math Library
- Initial Cardano section on OpenZeppelin Documentation

Security:

- Security audits of library components
- Continuous coverage and AI-Security Agent

Developer Enablement (included):

- Marketing & community activations
- Dedicated Technical Account Manager

Acceptance Criteria:

- All library code compiles against the chosen Cardano toolchain and passes CI with 100% test pass rate.
- 90% code coverage confirmed via automated test reporting.
- Access Control, Cryptographic Utilities, and DeFi Math components implemented and demonstrated with working examples.
- Architecture documents for the 3 Reference Implementations published and reviewed by pre-agreed counterparties.
- All code published in public GitHub repositories under MIT license (OpenZeppelin tooling under AGPL 3.0).

Milestone 2 - Liquid Staking and DeFi Enablers (Q2: Jan 2027 → Mar 2027)

Focus: First Reference Implementation and DeFi-enabling library components

Reference Implementations:

- RI 1 - Cardano Liquid Staking Protocol (working reference code, architecture documentation, demo front-end, threat model)

Contracts Library:

- Vaults
- DeFi Primitives
- Timelock and Pause
- Staking and Delegation Utilities

Security:

- Security audits of Milestone 1 deliverables (audit report published)
- Security reviews and penetration tests of RI 1
- Continuous coverage and AI-Security Agent

Developer Enablement (included):

- Marketing & community activations
- Dedicated Technical Account Manager
- Leadership meeting

Acceptance Criteria:

- Liquid Staking RI demonstrable on testnet: deposit, delegation across an SPO set, LST issuance and accrual, and redemption.
- Demo front-end functional for core Liquid Staking flows.
- Threat model identifies and documents eUTXO-specific attack vectors.
- Vaults, DeFi Primitives, Timelock/Pause, and Staking & Delegation components pass >90% code coverage with full CI integration.
- Educational materials on building DeFi protocols on Cardano published alongside the RI.
- M1 audit report published with all critical and high findings resolved.

Milestone 3 - Self-Repaying Loans and DevTools (Q3: Apr 2027 → Jun 2027)

Focus: Second Reference Implementation, governance/composability components, and developer tooling

Reference Implementations:

- RI 2 - Self-Repaying Loans Protocol (working reference code, architecture documentation, demo front-end, threat model)

Contracts Library:

- Governance
- Credentials & Claims
- Modular Account Abstraction & Multi-Sig
- OpenZeppelin Contracts Wizard, UI Builder, and Documentation for Cardano
- AI-Enhanced Development Tools for Cardano (MCP Server, AI Skills, Claude Plugin)

Security:

- Security audits of Milestone 2 deliverables (audit report published)
- Security reviews and penetration tests of RI 2
- Continuous coverage and AI-Security Agent

Developer Enablement (included):

- Marketing & community activations
- Dedicated Technical Account Manager

Acceptance Criteria:

- Self-Repaying Loans RI demonstrable on testnet: collateral deposit, synthetic-debt issuance, a harvest mechanism applying accrued yield to debt, and redemption.
- Demo front-end functional for core flows.
- Governance, Credentials & Claims, and Multi-Sig / Account Abstraction components pass >90% code coverage with full CI integration.
- Contracts Wizard generates valid Cardano contract code for token and vault configurations.
- AI tools (MCP Server) return correct library component usage patterns for Cardano-specific queries.
- Cardano Network section on OpenZeppelin Documentation is comprehensive for all released library components.
- M2 audit report published with all critical and high findings resolved.
- At least 1 independent Cardano developer has reviewed and provided feedback on the library components (written confirmation).

Milestone 4 - Tokenized MMF, Interop and Standards Leadership (Q4: Jul 2027 → Sep 2027)

Focus: Final Year 1 Reference Implementation, forward-looking standards leadership and interoperability, and the 12-month scope review.

Reference Implementations:

- RI 3 - Tokenized Money Market Fund (working reference code, architecture documentation, demo front-end, threat model)
- 12-Month Scope Review

Contracts Library:

- Vesting Contracts
- Standardized Messaging Gateway
- Cardano Contract ABI
- CIP-113 standard participation & design contributions
- Additional Standard Implementations (TBD)

Security:

- Security audits of Milestone 3 and 4 deliverables (audit reports published for Milestone 3 audits; established timeline for Milestone 4 audit publication)
- Security reviews and penetration tests of RI 3
- Continuous coverage and AI-Security Agent

Developer Enablement (included):

- Marketing & community activations
- Dedicated Technical Account Manager
- Leadership meeting

Acceptance Criteria:

- Tokenized MMF RI demonstrable on testnet: share issuance with KYC/AML credential verification, NAV-based subscription and redemption, dividend distribution, and settlement via USDCx or equivalent.
- Demo front-end functional for core flows.
- Vesting, Standardized Messaging Gateway, and Cardano Contract ABI components pass >90% code coverage with full CI integration.
- M3 audit reports published with all critical and high findings resolved. Critical and high findings in M4 audits reports addressed. Agreed publication timeline of M4 audit reports.
- 12-Month Scope Review completed with a Year 2 continuation proposal shared with the Cardano Foundation.

Funding

Total Funding Request: \$1,831,000 over 12 months. All pricing is denominated in **USD**. **Payments can be made in tokens or stablecoins, subject to review and approval by our finance team**, and can be scheduled over the duration of the agreement.

Payment Breakdown by Milestone

The engagement is delivered as an **initial kick-off payment followed by four quarterly, milestone-based installments**. Each milestone bundles the Reference Implementation, Contracts Library, and Security work delivered in that quarter into a single combined deliverable.

Milestone	Reference Implementations	Contracts Library	Security	Dev Enablement	Subtotal	% of Total
M0 (Kick-Off)	-	-	-	-	\$366,200	20%
M1 (Q1)	Research and Design	Access Control, Cryptography, DeFi Math	Audits, Continuous Coverage	Included	\$366,200	20%
M2 (Q2)	RI 1: Liquid Staking	Vaults, DeFi Primitives, Timelock/Pause, Staking & Delegation	Audits, Reviews, Pen Tests	Included	\$366,200	20%
M3 (Q3)	RI 2: Self-Repaying Loans	Governance, Credentials & Claims, Multi-Sig/AA, Wizard, AI Tools	Audits, Reviews, Pen Tests	Included	\$366,200	20%
M4 (Q4)	RI 3: Tokenized MMF, 12-mo Review	Vesting, Messaging Gateway, Contract ABI, CIP-113 design contributions, Additional Standards	Audits, Reviews, Pen Tests	Included	\$366,200	20%

Maintenance

All code produced under this grant will be published as open source: the Contracts Library and Reference Implementations under the MIT license, and OpenZeppelin tooling under the AGPL 3.0 license. **OpenZeppelin has a demonstrated track record of maintaining open-source contract libraries as long-term public goods across Ethereum (10+ years), Starknet, Stellar, Arbitrum, and others.**

Post-grant maintenance of the Cardano Contracts Library will follow the same model: OpenZeppelin will continue to maintain the library as part of its open-source portfolio. **Should the Cardano Foundation require a formal maintenance commitment beyond the grant period, this can be structured as a separate recurring maintenance grant.**

Open Source

All code produced under this grant will be released as open source:

- **Contracts Library:** Released under the MIT license, freely available for any Cardano developer to import, use, and build upon without restriction.
- **Reference Implementations:** Released under the MIT license, enabling Cardano builders to fork, adapt, and commercialize freely while contributing to ecosystem standardization.
- **OpenZeppelin Tooling:** Released under the AGPL 3.0 license, ensuring tooling derivatives remain open source and contribute back to the ecosystem.

All development will be conducted in public GitHub repositories under the OpenZeppelin organization, with full transparency into the development process, code reviews, and CI/CD pipeline.

Customer Testimonials

With a proven track record of delivering to leading ecosystems and protocols such as **Ethereum, Arbitrum, Starknet, Stellar, Midnight, Polkadot, and Uniswap**, OpenZeppelin is trusted by developers and foundation leaders.



Uniswap Ecosystem Stack

- ✓ OpenZeppelin Contracts Library for Hooks
- ✓ Gold Standard Audits for Uniswap v4
- ✓ Community Enablement

"OpenZeppelin's solutions simplify the implementation of complex features, making life significantly easier for developers working on advanced protocols like Uniswap v4."

Saucepoint

Product, Uniswap Foundation



Starknet Ecosystem Stack

- ✓ OpenZeppelin Contracts Library for Cairo
- ✓ SNIPs Standards Development
- ✓ Dev and security tooling
- ✓ Community Enablement
- ✓ Gold Standard Audits

"We enjoy working with OpenZeppelin to provide secure and reliable smart contracts that help developers build new amazing products on Starknet"

Derek Flossman

Product, Starknet Foundation



Arbitrum Ecosystem Stack

- ✓ OpenZeppelin Contracts Library for Stylus (Rust)

"We're thrilled to be collaborating with

- ✓ AIPs Standards Development
- ✓ Gold Standard Audits for Ecosystem Builders
- ✓ Defender cloud infrastructure support
- ✓ Community Enablement
- ✓ Security Council member for Arbitrum DAO

OpenZeppelin to help provide a secure environment for Stylus builders in the Arbitrum ecosystem."

Nina Rong

Head of Ecosystem Development, Arbitrum Foundation



Ethereum Ecosystem Stack

- ✓ Smart Contracts Library in Solidity
- ✓ Dev and security tooling
- ✓ EIPs & ERCs Standards Development
- ✓ Community Enablement
- ✓ Gold Standard Audits and Incident Response

"Collaborating with OpenZeppelin on our security audit was a productive and positive experience. We appreciated their thoroughness and attention to detail."

Yoav Weiss

Security, Ethereum Foundation



Polkadot Ecosystem Stack

- ✓ OpenZeppelin Library of Runtime Templates for Substrate
- ✓ PSPs Standards Development
- ✓ Dev and security tooling
- ✓ Community Enablement
- ✓ Gold Standard Audits

"We are ecstatic to join forces with @OpenZeppelin! This collaboration with the Polkadot community underscores our commitment to enhancing the development experience for Polkadot Parachain builders in a secure and seamless way."

Santiago Balaguer

Ecosystem, Parity

Stellar Ecosystem Stack

- ✓ OpenZeppelin Contracts Library for Soroban (Rust)
- ✓ SEPs Standards Development
- ✓ Dev and security tooling
- ✓ Community Enablement
- ✓ Gold Standard Audits

"By joining forces with OpenZeppelin, SDF is bringing the best-in-class smart contract development standards to the Stellar ecosystem. We're excited to work with the OpenZeppelin team to create the foundation for the next generation of applications built on Stellar."

Lindsay Poss

Stellar Development Foundation

Midnight Ecosystem Stack

- ✓ OpenZeppelin Contracts Library for Compact
- ✓ MIPs and MRCs Standards Development
- ✓ Dev and security tooling
- ✓ Community Enablement
- ✓ Gold Standard Audits

"With OpenZeppelin's instrumental support, we're helping ecosystem developers create new DApps and elevate existing ones to offer data protection while ensuring compliance in today's rapidly shifting landscape. This partnership transforms the security and optimization of applications across various blockchain ecosystems, paving the way for a seamless, private, multichain future"

Eran Barak

CEO, Midnight

Zama Ecosystem Stack

- ✓ OpenZeppelin Confidential Contracts Library for Zama's fhEVM
- ✓ Dev and security tooling
- ✓ Community Enablement
- ✓ Gold Standard Audits

"Our partnership with OpenZeppelin, whose smart contract standards secure 218 billions in digital assets, enables us to build confidential primitives with the security and auditability that institutions demand.

Together, we're creating the confidential computing layer that will unlock blockchain's next chapter."

Rand Hindi
CEO, Zama

Appendix

Detailed Contracts List

This table showcases the most important features and standards that are present in the solidity library together with high-profile implementations.

Features	Contract standards	Implementation Examples
Tokens	• ERC20 • ERC721 • ERC4626 • ERC1155 • ERC2981	• USDC token • BAYC NFT • AAVE vault
Security	• Pausable • Reentrancy Guard • Safe ERC20	• Ion Pool • AAVE V3 Periphery • Chainlink CCIP
Upgradeability	• Standard, Beacon, Transparent, UUPS Proxy • Clones	• AAVE V3 Core
Governance	• Governor • Vote • Timelock	• Arbitrum Governor
Access Control	• Ownable • Access Control • Access Manager	• Gelato
Cryptography	• ECDSA • EIP712 • Hashes • Merkle Proof • RSA • Signature Checker	• Uniswap Merkle Distributor
Meta Transactions	• ERC2771	• Gelato
Finance	• Vesting Wallet	• Arbitrum Governance
Utilities	• Data Structures (Checkpoints, Bitmaps, EnumerableMap) • Type extensions (Arrays, Strings) • Math • Multicall	• Biconomy
Cross-Chain	• ERC-7786 (Cross-Chain Messaging Gateway) • ERC-7802 (ERC20Bridgeable) • CrosschainLinked • ERC20Crosschain	• Axelar

Accounts

- ERC-4337 (Account, AccountERC7579)
 - ERC-7579 (Modular Account Modules)
 - ERC-7702 (SignerERC7702)
 - Signers: SignerECDSA, SignerP256, SignerRSA
 - ERC-7821 Batch Executor
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